

CHARLWOOD, United Kingdom

WMO: 037690

Lat: 51.144N

Lon: 0.229W

Elev: 223

StdP: 14.58

Time Zone: 0.00 (EUW)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	24.0	26.6	20.7	15.6	27.9	23.1	17.5	29.6	22.0	43.7	20.1	44.6	2.5	10	0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	82.1	67.5	78.0	65.3	74.6	63.6	68.7	79.6	66.6	75.6	64.9	72.2	7.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	92.5	72.6	63.2	87.6	69.9	61.9	83.4	68.1	33.0	79.9	31.3	75.6	30.0	72.2	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.5	16.3	14.4	DB	18.1	88.3	3.5	3.9	15.6	91.0	13.5	93.3	11.6	95.5	9.0	98.3	
			WB	17.6	72.5	3.5	2.3	15.0	74.2	13.0	75.5	11.0	76.9	8.5	78.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	40.3	40.4	43.7	48.7	53.7	59.6	63.3	62.2	57.8	52.1	45.3	41.4	(6)
	DBStd	9.82	6.06	5.76	5.43	5.37	5.19	4.57	4.47	4.23	4.49	5.43	5.97	6.65	(7)
	HDD50	1356	302	271	202	83	20	0	0	0	2	40	161	275	(8)
	HDD65	5263	765	688	660	488	350	171	88	110	220	399	591	733	(9)
	CDD50	1641	2	3	8	45	136	289	413	377	235	106	20	7	(10)
	CDD65	72	0	0	0	0	1	10	36	22	3	0	0	0	(11)
	CDH74	884	0	0	0	8	30	155	403	235	48	5	0	0	(12)
	CDH80	204	0	0	0	0	2	30	104	63	4	1	0	0	(13)
(14) Wind	WSAvg	6.3	7.3	7.2	7.4	6.4	6.3	6.1	5.8	5.4	5.4	6.0	6.1	6.8	(14)
(15) Precipitation	PrecAvg	30.0	3.1	2.0	2.2	2.1	2.0	2.2	1.9	2.2	2.7	3.0	3.2	3.2	(15)
	PrecMax	41.7	7.2	5.6	4.6	4.4	5.6	5.4	4.1	5.2	8.8	7.7	7.7	5.7	(16)
	PrecMin	20.0	0.6	0.3	0.2	0.2	0.0	0.3	0.4	0.0	0.5	0.1	0.6	0.6	(17)
	PrecStd	4.1	1.7	1.3	1.1	1.2	1.1	1.4	1.0	1.3	2.0	1.9	1.8	1.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.4	58.2	64.9	74.6	77.6	84.1	88.2	86.9	79.3	69.5	60.5	56.5	(19)
		MCWB	52.1	50.9	53.2	60.1	63.9	69.5	70.3	70.7	66.2	62.2	58.1	54.0	(20)
	2%	DB	53.0	53.4	59.2	67.9	73.2	78.1	82.8	80.2	74.1	65.0	58.3	54.7	(21)
		MCWB	51.1	49.5	51.3	56.8	61.9	65.6	67.3	67.4	64.0	59.8	55.8	52.7	(22)
	5%	DB	51.3	51.2	55.8	63.1	68.9	74.2	78.6	75.3	70.1	62.5	56.5	53.2	(23)
		MCWB	49.4	48.1	49.4	54.1	59.3	63.3	65.4	64.7	61.9	58.4	54.0	51.3	(24)
	10%	DB	49.8	49.4	52.8	59.5	64.9	70.7	74.4	71.7	67.2	60.6	54.8	51.7	(25)
		MCWB	47.8	46.8	48.2	51.9	56.6	61.7	63.7	62.6	60.1	57.4	52.6	49.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	52.3	54.3	61.2	66.1	70.6	71.8	71.6	68.1	63.3	58.8	54.7	(27)
		MCDB	53.7	55.2	60.3	72.4	74.6	82.5	85.6	84.9	76.4	67.5	59.9	55.7	(28)
	2%	WB	51.6	50.5	52.4	57.5	62.9	66.9	68.4	68.2	65.4	61.0	56.4	53.2	(29)
		MCDB	52.6	52.5	57.2	66.2	70.9	75.7	80.2	78.4	71.7	63.6	57.7	54.3	(30)
	5%	WB	49.8	48.8	50.8	55.0	60.1	64.5	66.5	65.7	63.4	59.6	54.6	51.8	(31)
		MCDB	51.0	50.6	54.8	61.8	67.5	72.3	76.5	72.8	68.1	61.7	56.1	53.0	(32)
	10%	WB	48.2	47.2	49.2	52.8	57.8	62.4	64.6	63.9	61.6	58.2	52.8	49.9	(33)
		MCDB	49.6	49.2	52.5	58.1	63.5	69.3	72.6	69.4	65.4	60.1	54.4	51.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	11.8	14.7	18.2	18.1	18.8	19.1	18.2	18.0	14.6	12.4	11.0	(35)
		MCDBR	10.7	14.5	20.5	25.6	24.9	26.1	27.4	24.8	23.5	15.9	12.4	9.4	(36)
		MCWBR	10.5	10.7	13.4	15.6	14.6	15.0	13.7	13.0	14.0	11.3	10.7	8.9	(37)
	5% WB	MCDBR	10.4	12.5	18.1	23.0	22.4	23.8	24.2	21.9	19.4	13.5	12.1	9.4	(38)
		MCWBR	10.5	10.2	12.8	14.5	13.7	14.5	13.0	12.4	12.5	10.5	11.1	9.4	(39)
(40) Clear-Sky Solar Irradiance	taub		0.329	0.345	0.382	0.401	0.405	0.405	0.400	0.393	0.376	0.359	0.333	0.319	(40)
	taud		2.372	2.377	2.269	2.240	2.272	2.297	2.317	2.358	2.398	2.423	2.417	2.362	(41)
	Ebn at Noon		216	246	256	266	270	270	269	266	257	238	214	200	(42)
	Edh at Noon		20	26	35	40	40	40	38	35	30	24	19	17	(43)
(44) All-Sky Solar Radiation	RadAvg		259	477	841	1325	1594	1766	1677	1399	1045	598	311	202	(44)
	RadStd		21	62	109	134	102	164	147	108	81	53	35	25	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (75 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page